HỌC VIỆN CÔNG NGHỆ BƯU CHÍNH VIỄN THÔNG

KHOA CÔNG NGHỆ THÔNG TIN 1

BÁO CÁO GIỮA KỲ

Xây dựng Chatbot hỏi đáp về lịch sử Việt Nam sử dụng GraphRAG

Lớp: E22HTTT

Tên: Nguyễn Đức Anh

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Người hướng dẫn: Kim Ngọc Bách

Hà Nội – 2026

BÁO CÁO TUẦN 4

How to Construct a Knowledge Graph with an LLM

The construction process

Typically, you would follow these steps to construct a knowledge graph from unstructured text using an LLM:

1. Gather the data
2. Chunk the data
3. Vectorize the data
4. Pass the data to an LLM to extract nodes and relationships
5. Use the output to generate the graph

Gather your data sources

The first step is to gather your unstructured data. The data can be in the form of text documents, PDFs, publicly available data, or any other source of information.

Depending on the format, you may need to reformat the data into a format (typically text) that the LLM can process.

The data sources should contain the information you want to include in your knowledge graph.

Chunk the data

The next step is to break down the data into right-sized parts. This process is known as chunking.

The size of the chunks depends on the LLM you are using, the complexity of the data, and what you want to extract from the data.

You may not need to chunk the data if the LLM can process the entire document at once and it fits your requirements.

Vectorize the data

Depending on your requirements for querying and searching the data, you may need to create vector embeddings. You can use any embedding model to create embeddings for each data chunk, but the same model must be used for all embeddings.

Placing these vectors into a Vector index allows you to perform semantic searches, similarity searches, and clustering on the data.

Extract nodes and relationships

The next step is to pass the unstructured text data to the LLM to extract the nodes and relationships.

You should provide a suitable prompt that will instruct the LLM to:

- Identify the entities in the text.
- Extract the relationships between the entities.
- Format the output so you can use it to generate the graph, for example, as JSON or another structured format.

Optionally, you may also provide additional context or constraints for the extraction, such as the type of entities or relationships you are interested in extracting.

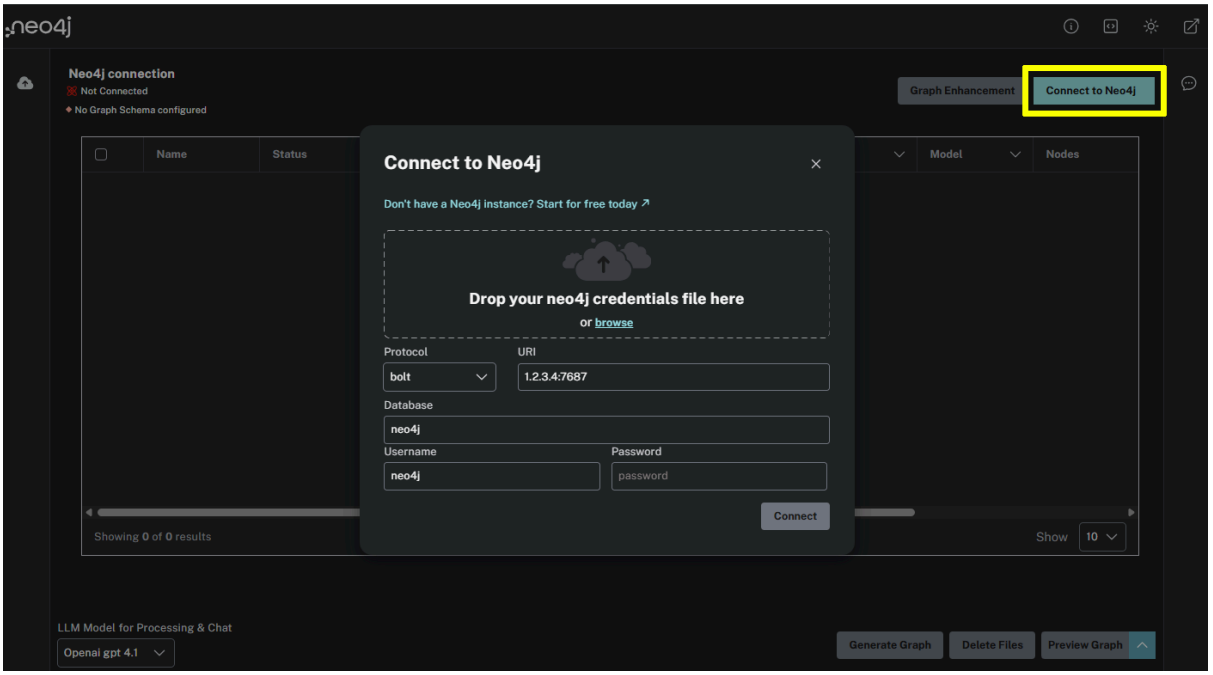
Generate the graph

Finally, you can use the output from the LLM to generate the graph by creating the nodes and relationships within Neo4j.

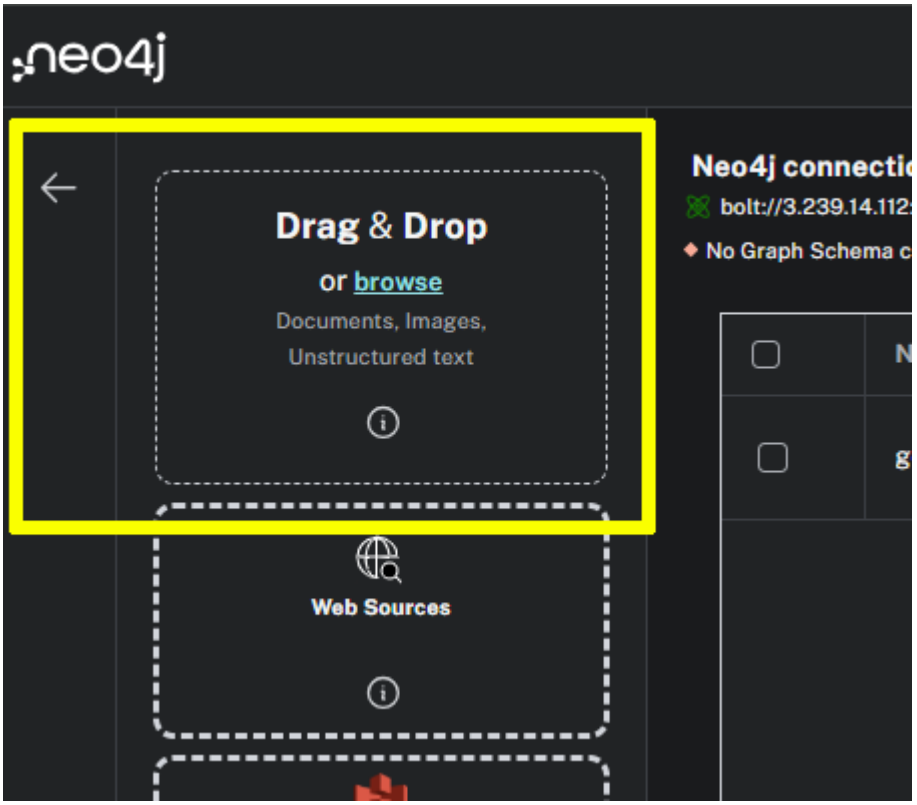
The entity and relationship types would become labels and relationship types in the graph. The names would be the node and relationship identifiers.

Connect to Neo4j

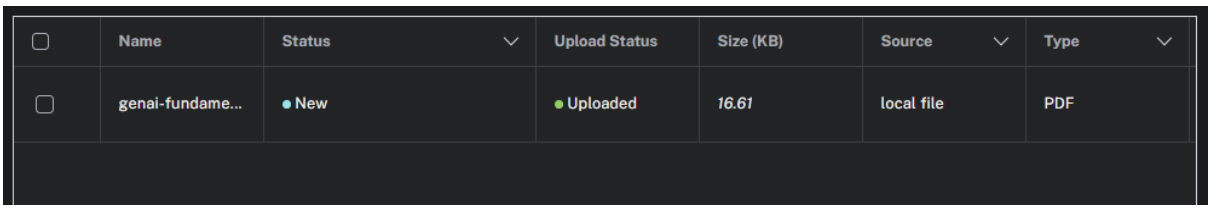
When you open the Neo4j LLM Graph Builder, you will be prompted to Connect to Neo4j. The LLM Graph Builder will create your knowledge graph in this database.



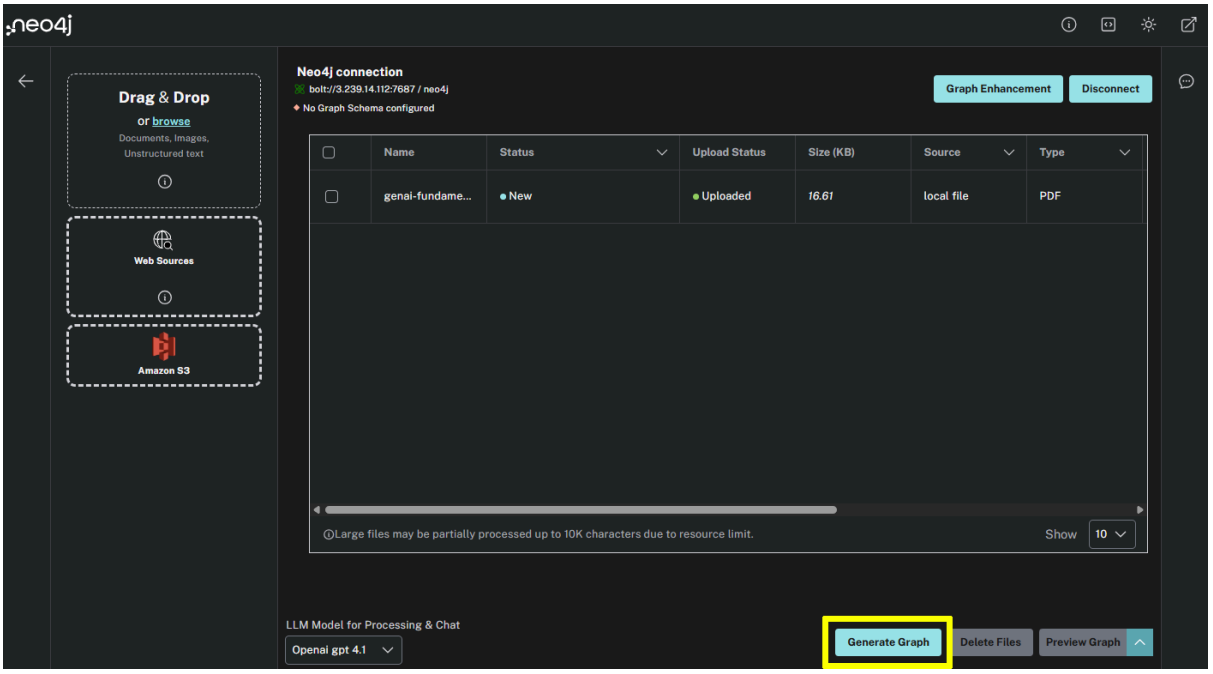
Upload the course content



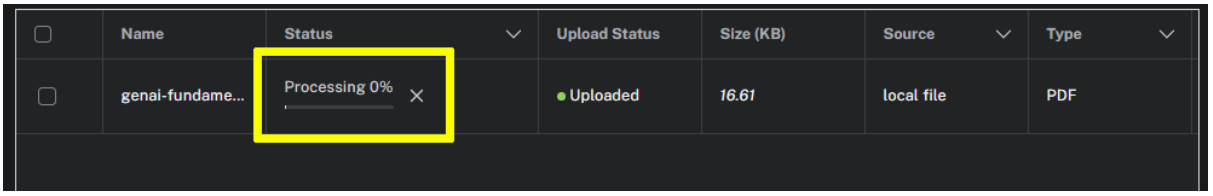
The file will uploaded and added to the list of documents.



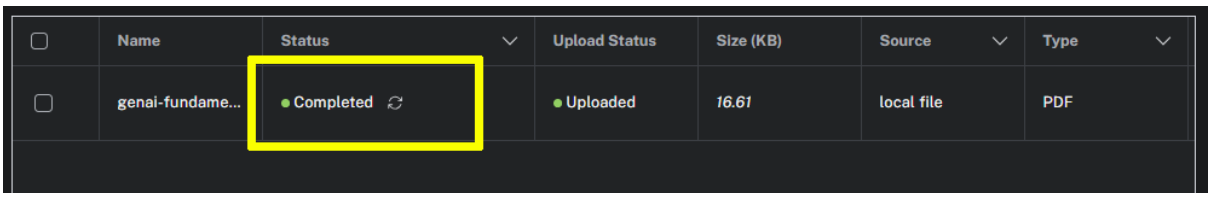
Once you have uploaded the document, you can create the knowledge graph using the Generate Graph button.



When generating the graph, the LLM Graph Builder runs the steps described in the previous lesson. The application chunks the data, creates vectors, extracts the nodes and relationships, and creates the graph.

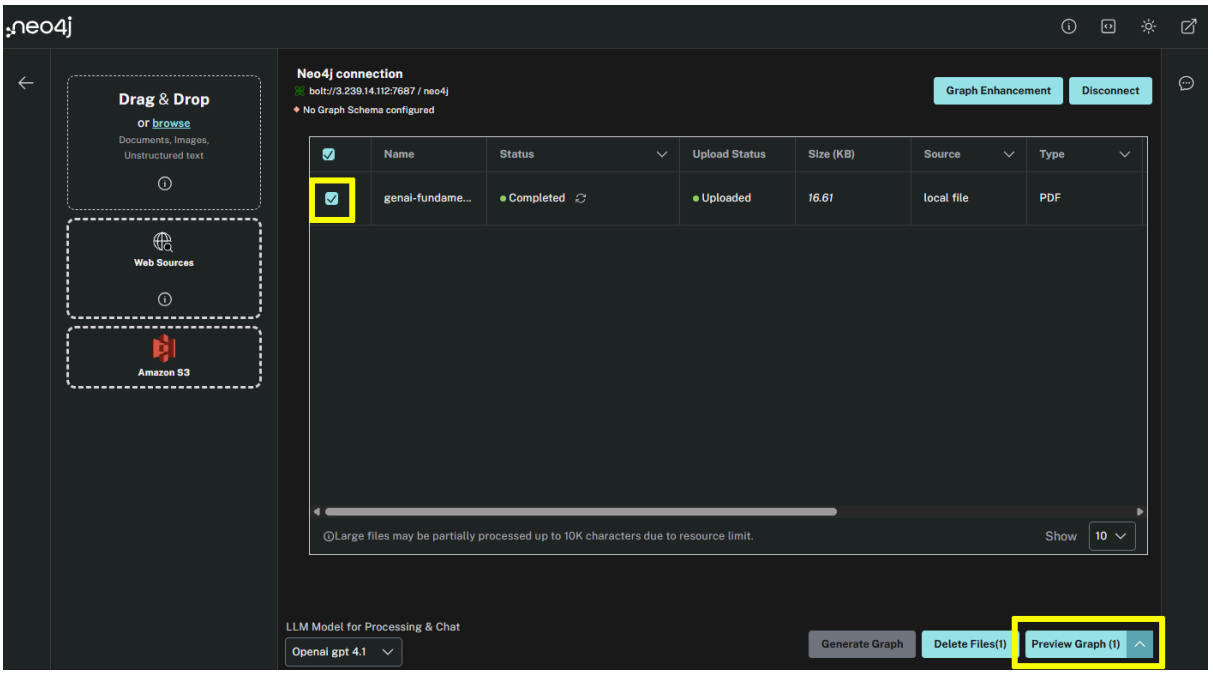


Once the document has completed processing, the knowledge graph will be available for exploration.

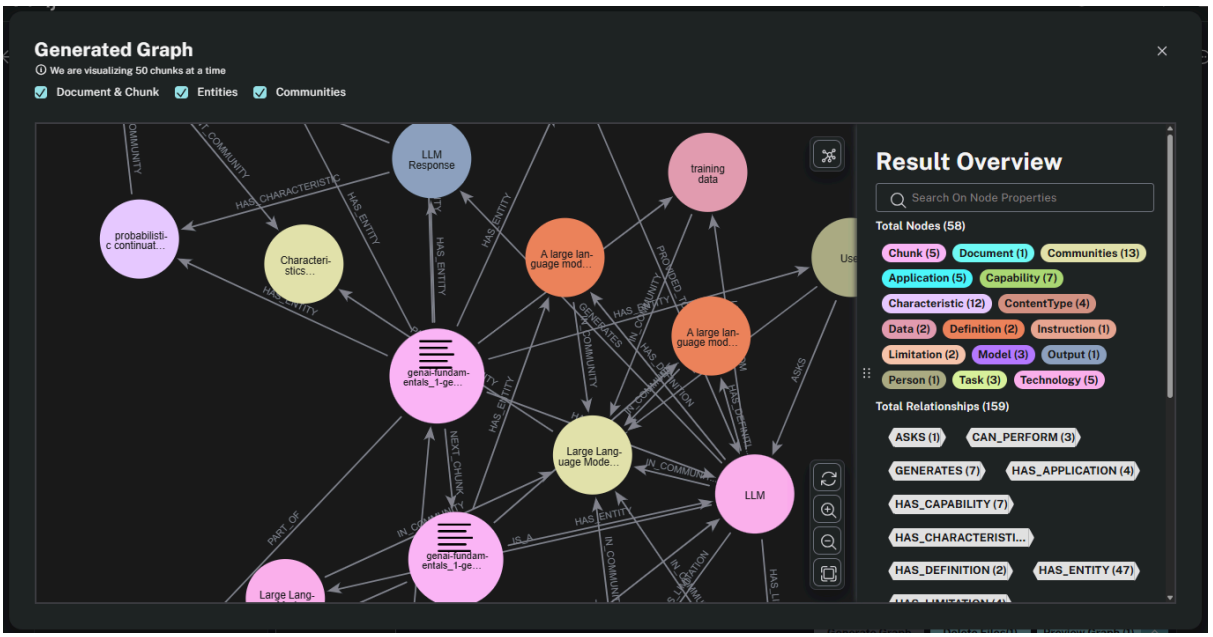


Preview

You can preview the graph for a single document by selecting the document and clicking the Preview Graph button.

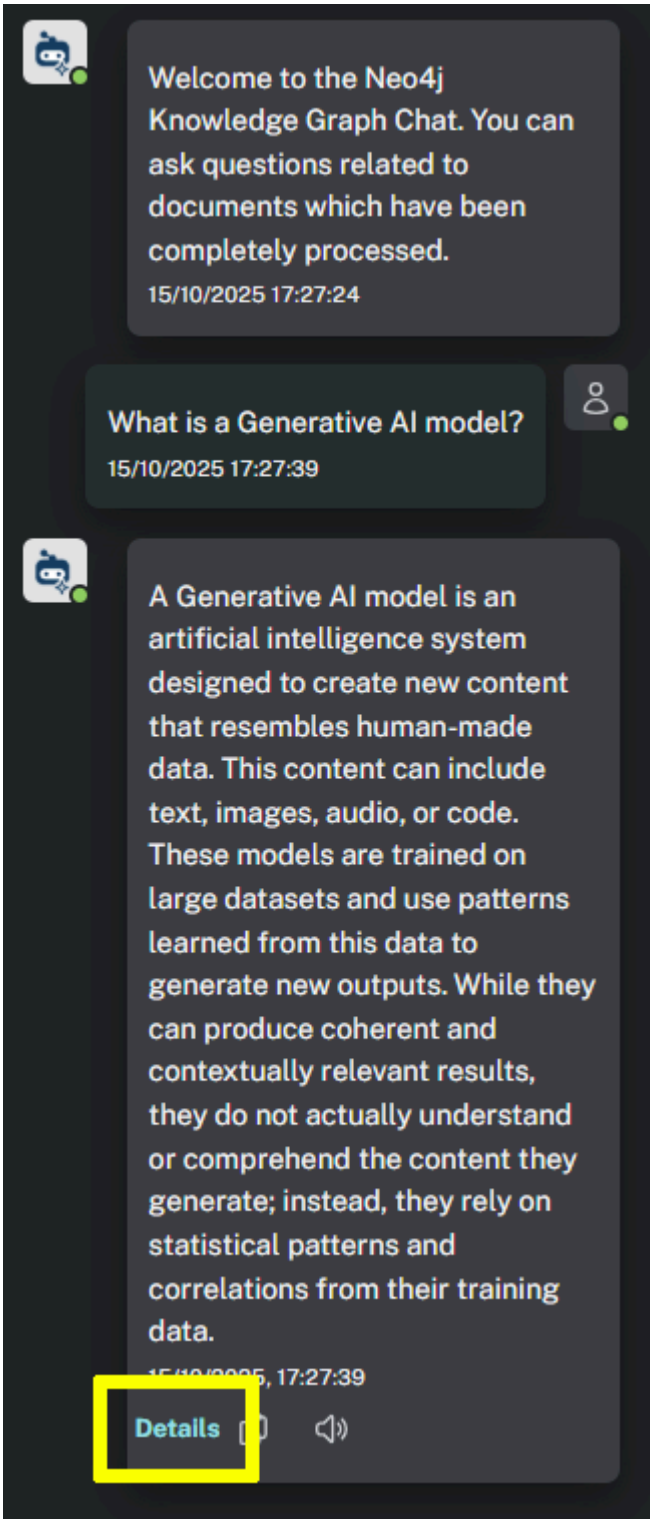


The preview shows the nodes and relationships extracted from the document.



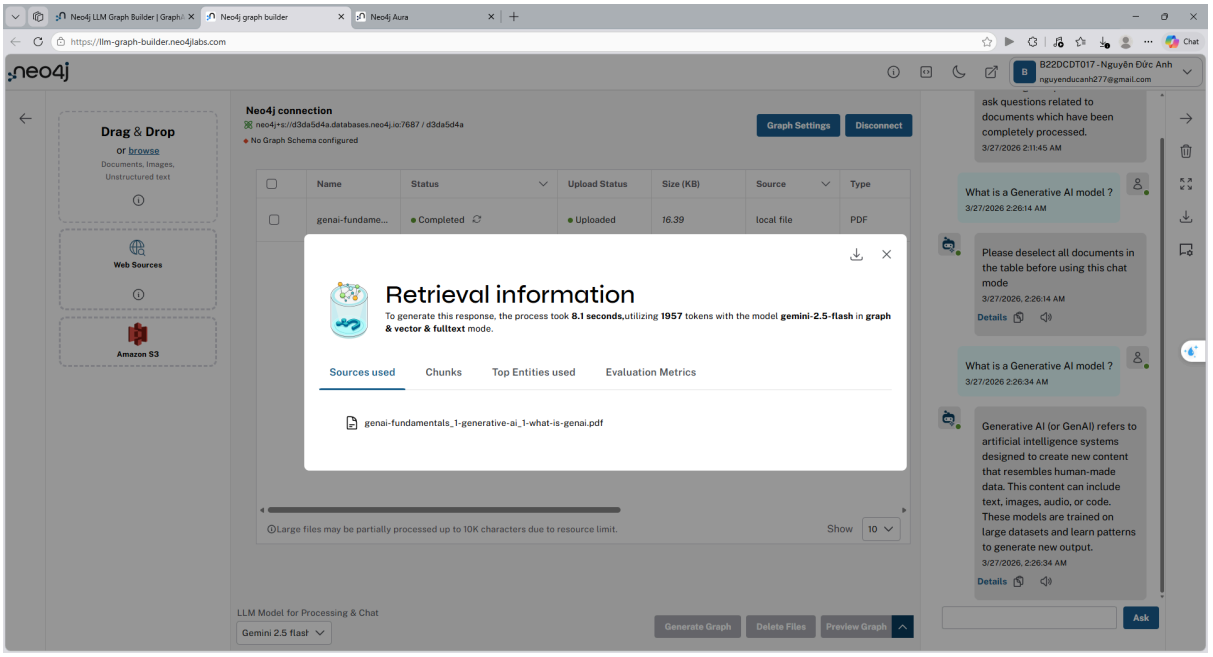
Query the Knowledge Graph

You can use the built-in chatbot to ask questions about the document's content.



The chatbot will use the knowledge graph to find the answer.

You can see the context used to answer the question in the Details.


Neo4j LLM Graph Builder | Graph
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or browse
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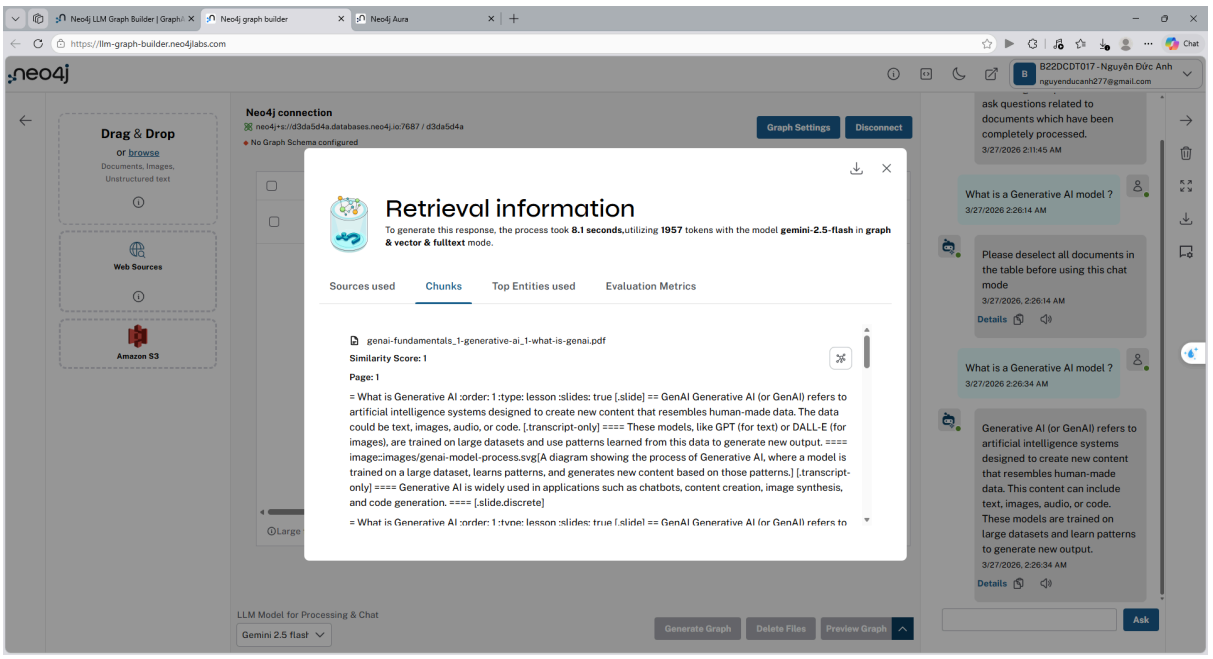
Neo4j connection
neo4j+s://d3da5d4a.databases.neo4j.io:7687 / d3da5d4a
No Graph Schema configured
Graph Settings
Disconnect
Name
Status
Upload Status
Size (KB)
Source
Type
genai-fundame...
Completed
Uploaded
16.39
local file
PDF
Retrieval information
To generate this response, the process took 8.1 seconds,utilizing 1957 tokens with the model gemini-2.5-flash in graph & vector & fulltext mode.
Sources used
Chunks
Top Entities used
Evaluation Metrics
genai-fundamentals_1-generative-ai_1-what-is-genai.pdf
Large files may be partially processed up to 10K characters due to resource limit.
Show
10
LLM Model for Processing & Chat
Gemini 2.5 flash
Generate Graph
Delete Files
Preview Graph
ask questions related to documents which have been completely processed.
3/27/2026 2:11:45 AM
What is a Generative AI model ?
3/27/2026 2:26:14 AM
Please deselect all documents in the table before using this chat mode
3/27/2026, 2:26:14 AM
Details
What is a Generative AI model ?
3/27/2026 2:26:34 AM
Generative AI (or GenAI) refers to artificial intelligence systems designed to create new content that resembles human-made data. This content can include text, images, audio, or code. These models are trained on large datasets and learn patterns to generate new output.
3/27/2026, 2:26:34 AM
Details
Ask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neo4j+s://d3da5d4a.databases.neo4j.io:7687 / d3da5d4a
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genai-fundamentals_1-generative-ai_1-what-is-genai.pdf
Similarity Score: 1
Page: 1
= What is Generative AI :order: 1 :type: lesson :slides: true [.slide] == GenAI Generative AI (or GenAI) refers to artificial intelligence systems designed to create new content that resembles human-made data. The data could be text, images, audio, or code. [.transcript-only] ==== These models, like GPT (for text) or DALL-E (for images), are trained on large datasets and use patterns learned from this data to generate new output. ==== image::images/genai-model-process.svg[A diagram showing the process of Generative AI, where a model is trained on a large dataset, learns patterns, and generates new content based on those patterns.] [.transcript-only] === Generative AI is widely used in applications such as chatbots, content creation, image synthesis, and code generation. ==== [.slide.discrete]
= What is Generative AI :order: 1 :type: lesson :slides: true [.slide] == GenAI Generative AI (or GenAI) refers to
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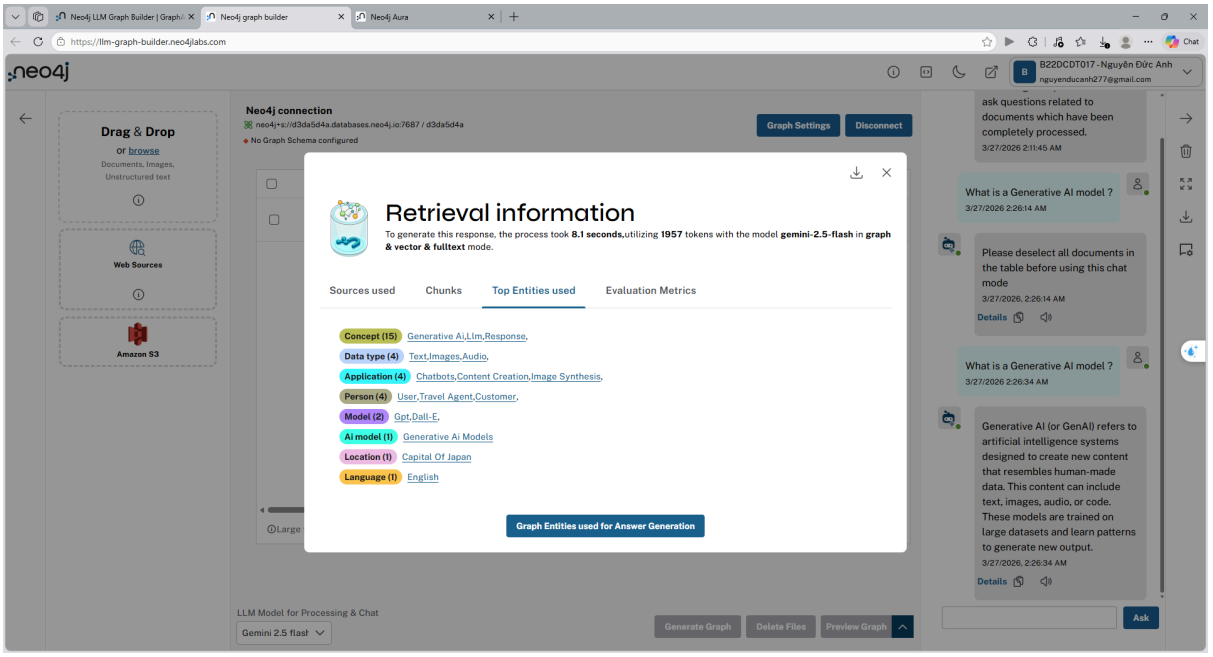
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Sources used
Chunks
Top Entities used
Evaluation Metrics
Concept (15) Generative Ai,Llm,Response,
Data type (4) Text,Images,Audio,
Application (4) Chatbots,Content Creation,Image Synthesis,
Person (4) User,Travel Agent,Customer,
Model (2) Gpt,Dall-E,
Ai model (1) Generative Ai Models
Location (1) Capital Of Japan
Language (1) English
Graph Entities used for Answer Generation
ask questions related to documents which have been completely processed.
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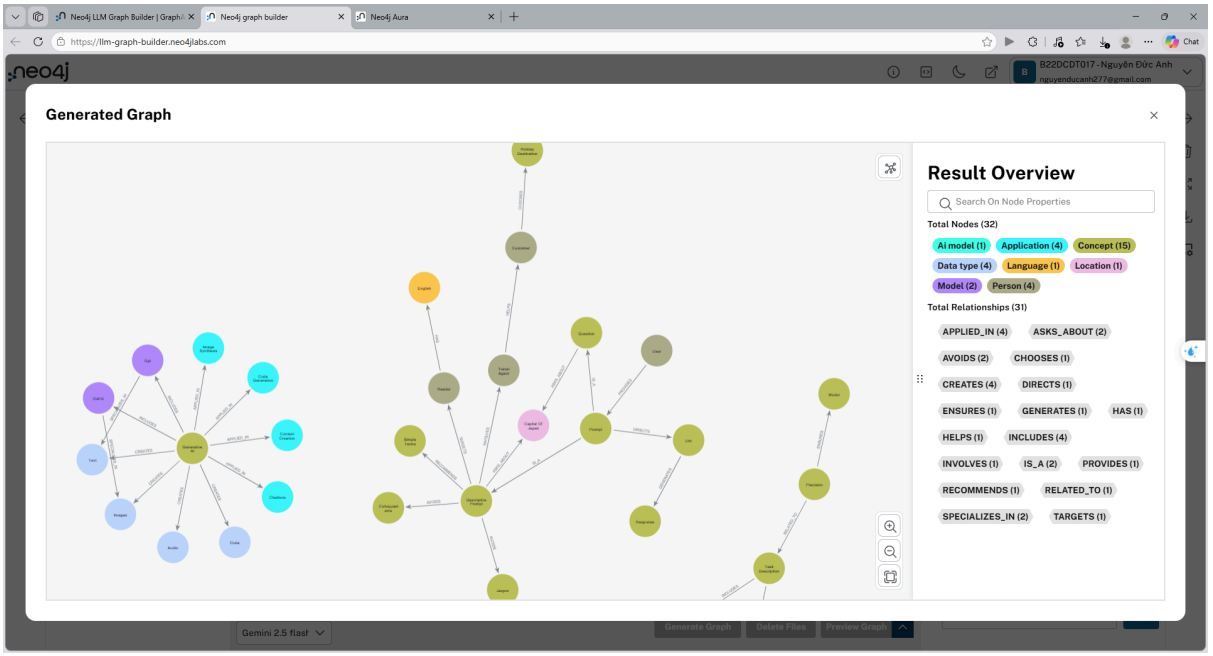
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neo4j
B22DCDT017 - Nguyễn Đức Anh
nguyenducanh277@gmail.com
Generated Graph
Result Overview
Search On Node Properties
Total Nodes (32)
Ai model (1)
Application (4)
Concept (15)
Data type (4)
Language (1)
Location (1)
Model (2)
Person (4)
Total Relationships (31)
APPLIED_IN (4)
ASKS_ABOUT (2)
AVOIDS (2)
CHOOSES (1)
CREATES (4)
DIRECTS (1)
ENSURES (1)
GENERATES (1)
HAS (1)
HELPS (1)
INCLUDES (4)
INVOLVES (1)
IS_A (2)
PROVIDES (1)
RECOMMENDS (1)
RELATED_TO (1)
SPECIALIZES_IN (2)
TARGETS (1)
Gemini 2.5 flash
Generate Graph
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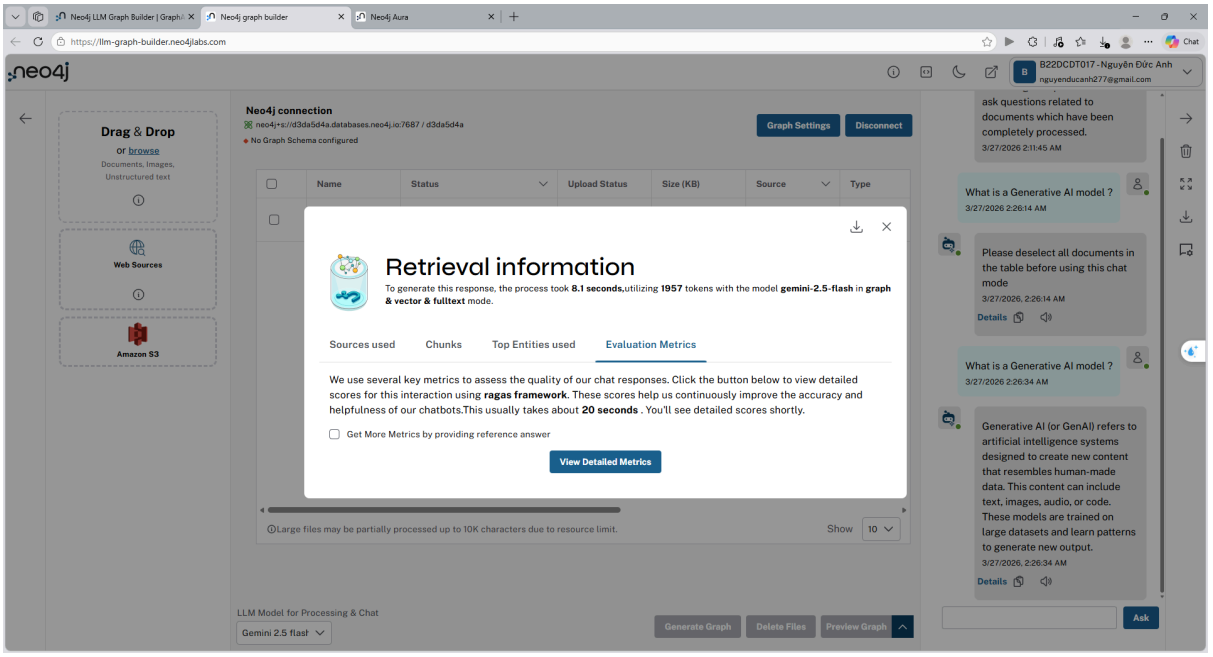
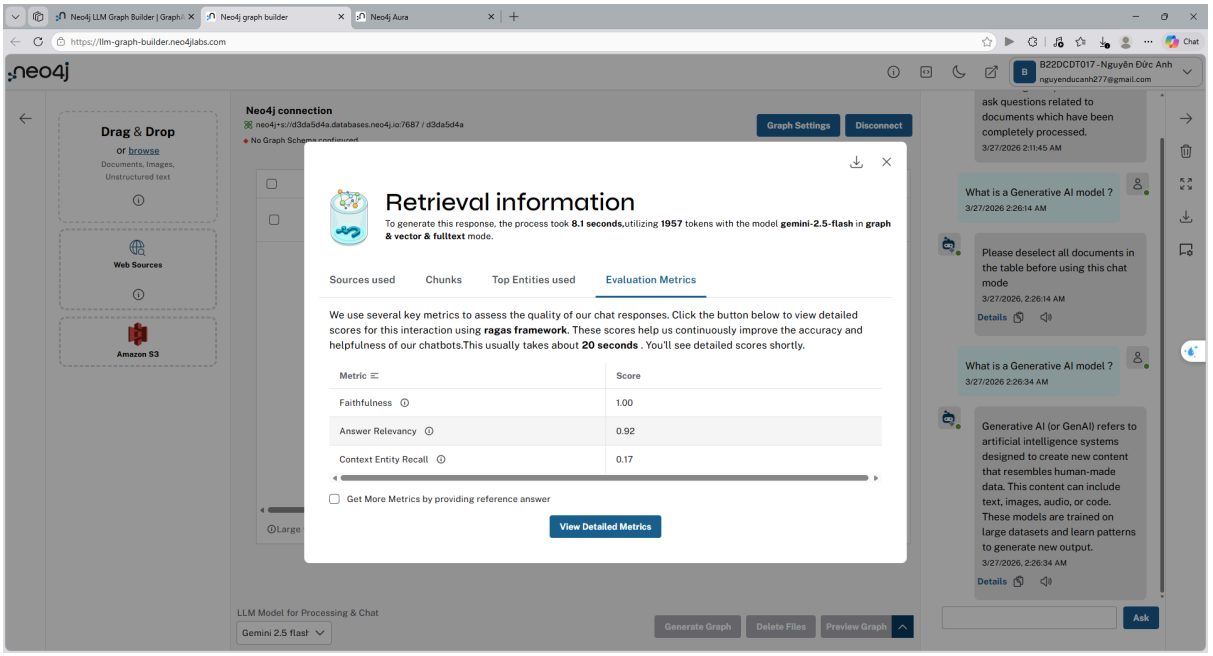
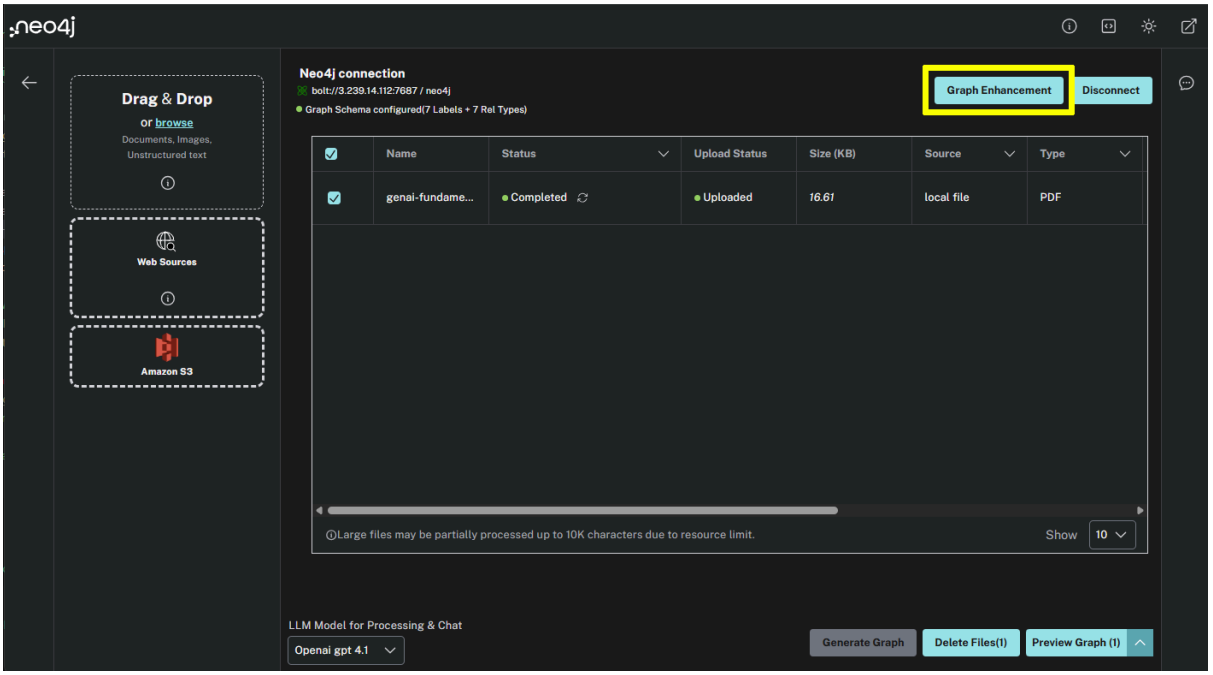


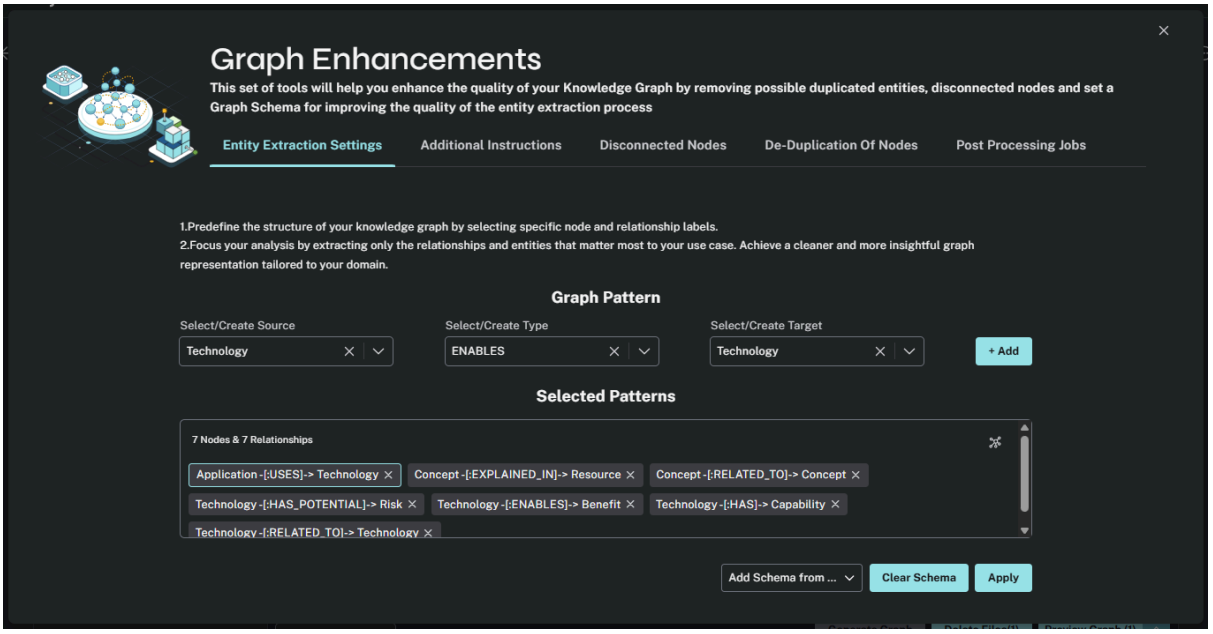


Define a schema

The schema is a set of graph patterns of node labels and relationship types ((node)-[RELATIONSHIP]→(node)), you want to identify within the text.

To define a schema, click the Graph Enhancement button.





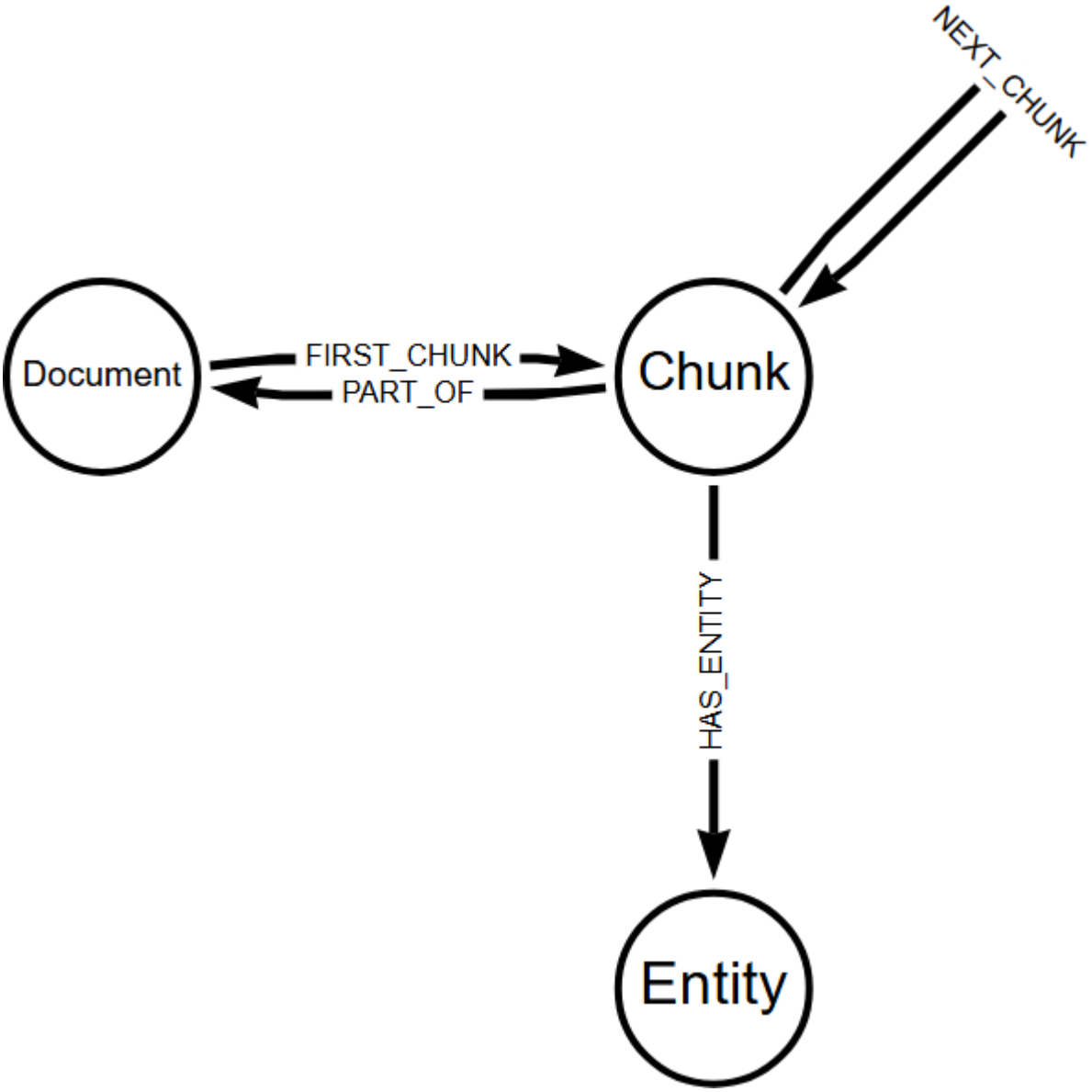
Customization

Through the Graph Enhancement feature, you can also customize:

- The prompt used to extract entities and relationships.
- The chunking strategy used to split the document into smaller pieces.
- Deleting disconnected nodes from the graph.
- De-duplicating nodes.
- Fine tuning any post processing of the graph.

Model

The structure created by the LLM Graph Builder is a good template for a knowledge graph that you can use in your projects.



1. Each Document node represents a document or text source uploaded to the LLM Graph Builder.
2. The Document is split into Chunk nodes, identified by FIRST_CHUNK and PART_OF relationships.
3. The LLM processes the chunks and Entity nodes extracted. The graph connects the Entity nodes to the Chunk nodes using the HAS_ENTITY relationship.

The LLM may also extract relationships between the entities, and the graph will hold relationships between the Entity nodes.

This model allows you to relate the entities within the knowledge graph to the source documents they refer to.

Explore

```
MATCH (d:Document

    {fileName:'genai-fundamentals_1-generative-ai_1-what-is-genai.pdf'}

    )

MATCH (d)<-[:PART_OF]-(c:Chunk)-[:HAS_ENTITY]->(e)

RETURN e
```

Running this query will return all the entities extracted from the document genai-fundamentals_1-generative-ai_1-what-is-genai.pdf

```
MATCH (d:Document
  {fileName: 'genai-fundamentals_1-generative-ai_1-what-is-genai.pdf'}
  )
MATCH (d)<-[:PART_OF]-(c:Chunk)-[:HAS_ENTITY]->(e)
RETURN labels(e) as labels, e.id as entity
```

You can modify the query to return the label and ID of the entities.

The results provide detail of the entities extracted from the document.

labels	entity
["Capability"]	"content creation"
["Technology", "Concept"]	"Generative A"
["Technology"]	"GPT"
["Risk"]	"lack of understanding"
…	

The Entity node label is a generic label applied to all entities to manage indexing and querying.

By matching a path from the document to the entities, you can see how the entities relate to the document.

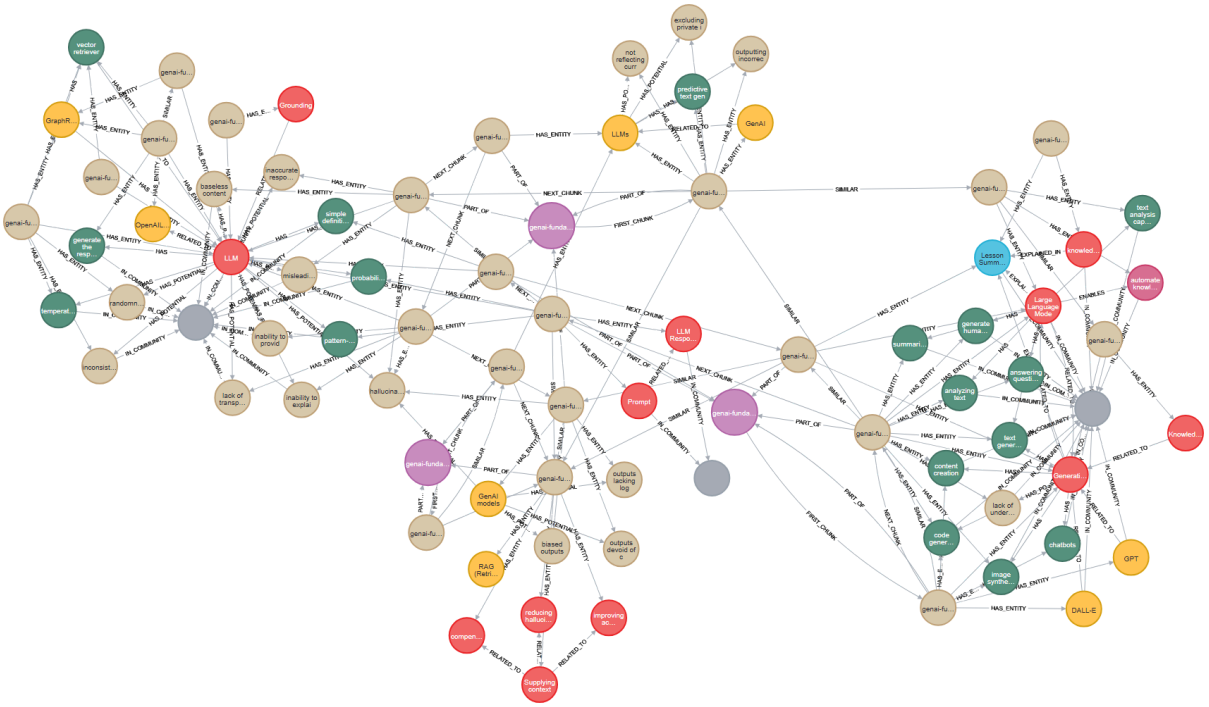
```
MATCH (d:Document

    {fileName:'genai-fundamentals_1-generative-ai_1-what-is-genai.pdf'}

    )

MATCH p = (d)-[*0..3]-(e)

RETURN p
```

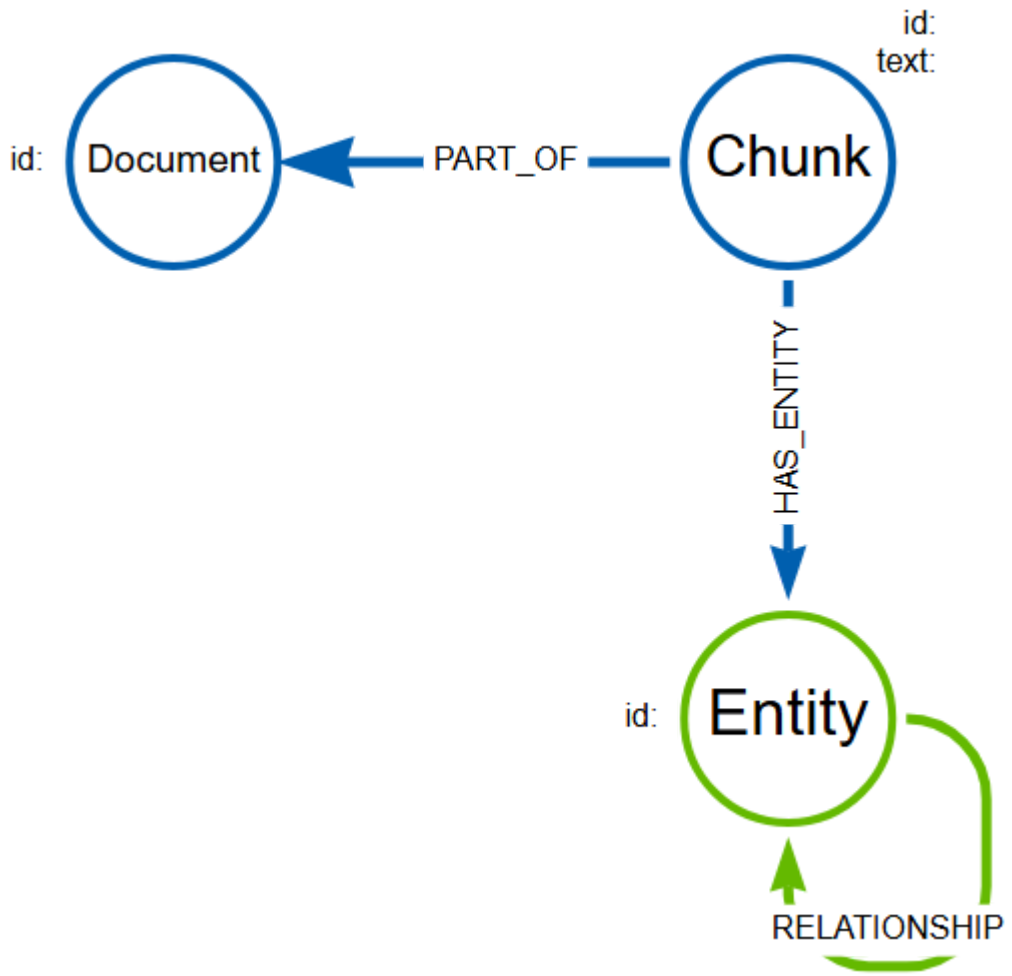


Querying using Cypher

The graph generated by an LLM may not conform to a strict schema (depending on how you configured the LLM). The lack of a strict schema changes how you query the graph. You may not be looking to find a specific node or relationship, but rather a pattern or structure in the graph.

The data structure of the graph contains both structured and unstructured elements.

Structured elements include the document and chunks. Unstructured elements include the entities and relationships extracted from the text.



You can use the structured elements to filter the graph and focus on specific parts of the graph. The unstructured elements allow you to explore the relationships between entities in the text.

Entities

The entities in the graph allow you to understand what the text is about.

For example, you can find the most mentioned topics in the graph by counting the number of times a node label (or entity) appears in the graph.

```
MATCH ()-[:HAS_ENTITY]->(e)

RETURN labels(e) as labels, count(e) as nodes

ORDER BY nodes DESC
```

A knowledge graph created from Neo4j & LLM Fundamentals course content contained the following entities:

labels	nodes
[Capability]	189
[Technology]	131
[Concept]	50
[Benefit]	44
[Risk]	35
["Technology", "Concept"]	31

You can drill down into the entity id to gain insights into the content. For example, you can find the most mentioned Technology.

```
MATCH ()-[r:HAS_ENTITY]->(e:Technology)

RETURN e.id AS entityId, count(r) AS mentions

ORDER BY mentions DESC
```

The course is about using Neo4j and LLMs with LangChain and OpenAI - this is reflected in the most mentioned technologies:

entityId	mentions
Neo4J	10
vector search	10
LLM	9
GraphRAG	9
retriever	7

Relationships

The relationships in the graph allow you to understand how entities are connected and the context of the text.

This Cypher query matches all the entities between the chunks and the relationships between them:

```
MATCH ()-[:HAS_ENTITY]->(entity)-[r]->(other)<-[:HAS_ENTITY]-()

RETURN DISTINCT type(r) AS relType, count(r) AS number

ORDER BY number DESC
```

relType	number
HAS	1242
RELATED_TO	853
USES	290
ENABLES	178
HAS_POTENTIAL	158
EXPLAINED_IN	51

You can build statements about the content by using the entity and relationship data, for example:

Technology Neo4J HAS Capability Vector search

This Cypher query uses the entity labels, id, and relationship type to create these statements:

```
MATCH ()-[:HAS_ENTITY]->(entity)-[r]->(other)<-[:HAS_ENTITY]-()

RETURN DISTINCT

    labels(entity), entity.id, type(r), labels(other), other.id
```

labels(entity)	entity.id	type(r)	labels(other)	other.id
[Technology]	GenAI Frameworks	HAS	[Capability]	Retrieval-augmented generation
[Application]	GenAI application development	USES	[Technology]	Neo4j
[Technology]	GenAI Frameworks	RELATED_TO	[Technology]	Neo4j
[Technology]	GraphRAG for Python	HAS	[Capability]	Knowledge graph construction
[Technology]	GraphRAG pipeline	ENABLES	[Benefit]	more accurate responses
[Skill]	Vector Indexes	USE_IN	[Technology]	Neo4J
[Technology]	Pip	INSTALLS	[Technology]	Youtube-Search

This structured output from the content allows you to understand the relationships between entities in the text.